Chandramauli Chakraborty

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Statistics PhD researcher working on **reinforcement learning from human feedback (RLHF)**, **Nash learning from human feedback (NLHF)**, and **multi-turn reinforcement learning**. Research interests include LLM post-training, AI agent alignment, robust evaluation, and multimodal LLMs.

EDUCATION

University of Chicago — PhD in Statistics 2024–present
Advisor: Prof. Cong Ma

Indian Statistical Institute, Kolkata — Master of Statistics 2022–2024
First Division with Distinction

Indian Statistical Institute, Kolkata — Bachelor of Statistics 2019–2022
First Division with Distinction

RESEARCH EXPERIENCE

Doctoral Research — University of Chicago 2024–present

- **RLHF and multi-turn RL:** Investigating reinforcement learning from human feedback in multi-turn reinforcement learning settings with Prof. Cong Ma.
- **Nash learning from human feedback (NLHF):** Studying preference-based policy learning through game-theoretic formulations and regularized minimax objectives.
- **Learning from demonstrations:** Established matching sample-complexity bounds under standard and robust pass@k evaluation with imperfect demonstrators; developed a greedy multiplicative-weights learner.
- **Multi-response evaluation:** Proved that pass@k ($k \geq 2$) improves worst-case accuracy dependence from $1/\varepsilon^2$ to $1/\varepsilon$ in a finite reward-class model with arbitrary demonstrators and no reward feedback.
- **Synthetic data and model collapse:** Studying how iterative training on synthetic data affects generative distributions and downstream classification, including when discriminative performance can persist.
- **Gaussian graphical models:** Investigating structure learning with Prof. Frederic Koehler, aiming for polynomial-time algorithms with improved sample complexity.

SELECTED PAPERS

The Statistical Benefits of Multiple Responses for Learning from Demonstrations
C. Chakraborty and C. Ma. Preprint; submitted to ICLR 2027.

Generative Collapse Does Not Imply Discriminative Collapse
C. Chakraborty*, J. Lee*, and C. Ma. Submitted to ICLR 2027.

Clustering High-dimensional Data with Ordered Weighted ℓ_1 Regularization
C. Chakraborty*, S. Paul*, S. Chakraborty, and S. Das. **AISTATS 2023**.

* Joint authorship.

ADDITIONAL RESEARCH AND PROJECTS

Generative Modeling Research Jun–Aug 2023
With Prof. Larry Wasserman and Prof. Arun Kumar Kuchibhotla

- Studied theoretical aspects of rectified flow and generative modeling.

Big Data Summer Institute — University of Michigan Jun–Aug 2022

- Used Python and spatial epidemiology to model county-level deficits in reported STI cases during the pandemic, in a project on reduced surveillance testing.

Visual Recognition Project — Indian Statistical Institute Spring 2022

- Built painting genre/style classifiers using hand-engineered visual features and clustering; developed methods to analyze paintings outside the genres used for training.

TECHNICAL SKILLS AND HONORS

Programming: Python, R, C/C++, SQL, shell scripting, LaTeX.
Methods: Statistical learning, minimax analysis, online learning, optimization, clustering.
Honors: Neubauer Fellowship, University of Chicago (2024); ACM India–IARCS Travel Grant (2023).